

Professor Jie Lu, Editor-in-Chief, *Knowledge-Based Systems*

Distinguished Professor and Director, Centre for Artificial Intelligence, University of Technology Sydney

Subject: Submission of Research Article: “PEM-CAN: Parameter-Efficient Multimodal Fusion for Diabetic Neuropathy Screening”

Dear Professor Lu and Editorial Board Members,

I am pleased to submit our original research manuscript entitled “**PEM-CAN: Parameter-Efficient Multimodal Fusion for Diabetic Neuropathy Screening**” for consideration for publication in *Knowledge-Based Systems*.

Motivation and Focus: Diabetic autonomic neuropathy (DAN) is a silent, lethal microvascular complication that doubles cardiovascular mortality. Early non-invasive screening requires unifying complementary physiological windows: ocular microvascular architecture (2D retinal fundus photography) and dynamic metabolic fluctuations (1D continuous glucose monitoring and wearable actigraphy). However, deep multimodal models face three fundamental engineering bottlenecks: (1) quadratic memory expansion $\mathcal{O}((M + L)^2)$ over longitudinal time series, (2) visual modality dominance that suppresses low-amplitude autonomic signals, and (3) severe telemetry packet loss during ambulatory surveillance.

Core Contributions directly aligned with *Knowledge-Based Systems*:

- **Stiefel-Manifold Orthogonal Low-Rank Adaptation:** By freezing pre-trained vision foundation backbones (ViT-B/16, 86.6M parameters) and constraining cross-attention projections to rank $r = 8$ factor matrices with a Stiefel orthogonality regularizer ($\mathcal{L}_{\text{orth}} = \|AA^T - I_r\|_F^2$), PEM-CAN mitigates modality collapse and updates only **1.75% of parameters** (2.60 M), reducing training VRAM from 32.4 GB to 8.4 GB.
- **State-of-the-Art Diagnostic Performance:** Evaluated on a clinical cohort statistically mapped to the NIH Bridge2AI AI-READI Type 2 Diabetes protocol ($N = 2,840$), PEM-CAN achieves **0.934 AUROC** [95% CI: 0.904, 0.960], **0.795 AUPRC**, and **0.038 ECE** at 90.5% accuracy, within 2.3% of the Bayes optimal oracle ceiling (0.957 AUROC).
- **Missing-Modality Resilience:** Under 50% contiguous telemetry packet dropout, PEM-CAN retains **0.861 AUROC** (92.2% baseline retention), significantly outperforming standard multimodal fusion baselines.
- **Distributed Federated Scaling and Edge Deployment:** Under 50-node federated training (FFA-LoRA), PEM-CAN slashes client communication overhead by **98.25%** (10.4 MB/round), reaching convergence $2.3\times$ faster. On embedded edge hardware (NVIDIA Jetson Orin Nano), the model executes in **20.2 ms** with a compact 1.18 GB FP16 footprint.

Author Declarations:

1. **Originality:** This manuscript is original, has not been published elsewhere, and is not under consideration by any other journal.
2. **Conflict of Interest:** The author declares no competing financial or personal interests.
3. **Generative AI Usage:** AI assistance was used strictly for grammatical polishing under full author oversight and responsibility.
4. **Open Science and Reproducibility:** Code and model configurations are public at <https://github.com/mahmoudsajjadi/PEM-CAN>.

Thank you for your consideration. I look forward to the reviewers’ constructive feedback.

Sincerely,

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